

A dark blue vertical bar is on the left. A blue arrow points right from it, containing the date.

5/8/2026

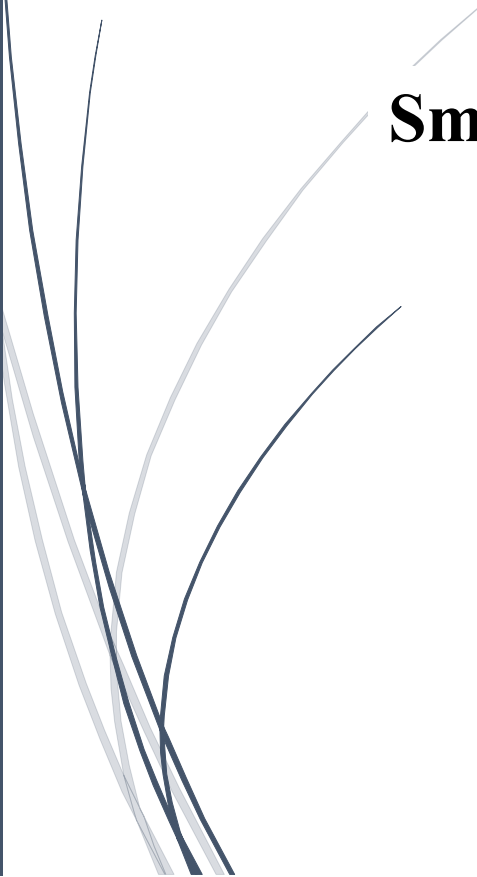
Lab Project

Machine Learning

Name: Hamza

Roll No: BSCS/562

Section: F

Several thin, curved lines in shades of blue and grey originate from the left side and curve upwards and to the right.

Smart Healthcare Prediction



Lahore Garrison University

Complex Computing Problem

Department of Computer Sciences

Instructor's Name: Sir Khurram Yousaf Tahseen

Program/Semester/: BSCS-6th

Section: F

Course Code: CSE6505

Course Title: Machine Learning

Time Allowed: 4 Weeks

Max Marks: 30

SUBJECTIVE TYPE (B)

The semester project has been designed to enable students to solve complex computing problems. Specifically, this project focuses on addressing the following characteristics of such problems in the context of Machine Learning:

WP-1	WP-3	WP-7	
Depth of Knowledge required of these fields. • Stats & Probability • Linear Algebra • Python Programming Skills • Web Development	Depth of Analysis required	Interdependence	
CLO Mapping with GA/PLO			
SR #	CLO	PLO	BT Levels
CLO 1	Understand machine learning algorithms, tools, and techniques.	PLO2	C = 2,
CLO2	Apply supervised and unsupervised learning, techniques for classification, regression.	PLO4	C = 3,
CLO5	Construct real world machine learning solutions using supervised and unsupervised algorithms with modern machine learning tools and libraries.	PLO 5	P-4

CCP Statement:

This project focuses on developing a **Smart Healthcare Prediction System for Diabetes Risk Assessment** using clinical and lifestyle data. The system applies ensemble machine learning techniques such as Random Forest, Gradient Boosting, AdaBoost, and Voting Classifier to improve prediction accuracy.

Unlike traditional binary prediction systems, this solution classifies patients into **three categories: Non-Diabetic, Pre-Diabetic, and Diabetic**, enabling early intervention. The system also provides **basic health recommendations** based on prediction results, supporting healthcare professionals in decision-making and improving patient outcomes.

Problem Identification:

Diabetes is a rapidly growing global health concern that can lead to severe complications such as heart disease, kidney failure, and vision loss if not detected early. However, accurate prediction is challenging due to **complex, noisy, and imbalanced medical datasets**.

Machine learning, particularly **ensemble learning methods**, has proven effective in identifying hidden patterns in healthcare data and improving prediction accuracy compared to traditional models. These techniques combine multiple models to produce more reliable and robust results.

Dataset:

Diabetes Prediction

<https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset?resource=download>

Deliverables:

Week	Deliverable	Description
Week 1	Problem Understanding	Literature review and understanding DIABETES dataset and project objectives
Week 2	Dataset Collection	Obtain Diabetes dataset and analyze dataset structure
Week 3	Exploratory Data Analysis	Visualize data, identify patterns, correlations, and missing values

Week	Deliverable	Description
Week 4	Data Preprocessing	Handle missing values, encode categorical features, and normalize data

Marks Distribution:

Total Marks = 30

- Feature Computing: Understand the dataset using Exploratory Data Analysis (EDA). Perform preprocessing including data cleaning, handling missing values, normalization, and feature encoding to prepare the dataset for machine learning models. **(6 marks – CLO 1)**
- Model Development: Implement ensemble machine learning algorithms such as Random Forest, Gradient Boosting, AdaBoost, and Voting/Stacking classifiers to build a predictive model for Diabetes using the dataset. **(6 marks – CLO 5)**
- Model Evaluation: Evaluate the performance of each model using appropriate metrics including accuracy, precision, recall, F1-score, and confusion matrix to assess prediction effectiveness. **(6 marks – CLO 2)**
- Model Comparison: Compare the performance of at least four ensemble models to determine the most accurate and reliable algorithm for Diabetes prediction. **(6 marks – CLO 2)**
- Documentation: Document the complete project pipeline including dataset description, preprocessing steps, model development, evaluation results, and system workflow, ensuring reproducibility and clarity of the machine learning solution. **(6 marks – CLO 1)**

Overview:

The project develops a predictive system to identify patients at risk of Diabetes using clinical data from a publicly available dataset. The data undergoes preprocessing steps such as cleaning, handling missing values, normalization, and feature encoding to prepare it for machine learning. Ensemble learning algorithms including Random Forest, Gradient Boosting, AdaBoost, and Voting classifiers are used to train models that analyze clinical features and detect patterns indicating DIABETES. Model performance is evaluated using accuracy, precision, recall, F1-score, and confusion matrices to identify the most reliable predictor.

The selected model is integrated into a web-based interface where users can input patient parameters such as blood pressure, hemoglobin level, and glucose level. Once submitted, the backend processes the data through the trained model and displays the prediction directly on the web interface, showing whether the patient is at risk of DIABETES. This end-to-end workflow—from data preprocessing and model training to real-time web-based prediction demonstrates how machine learning can enable early DIABETES detection, support healthcare decision-making, and provide a user-friendly system for medical professionals.

Reference Paper:

<https://www.sciencedirect.com/science/article/abs/pii/S1746809423008017>

❖ Working App Screen Shots:

The image displays two screenshots of a web application titled "Diabetes Risk Assessment".

Top Screenshot: The application is running in a browser window. The title bar shows "Diabetes Risk Assessment". The address bar shows "127.0.0.1:5000". The main content area features a heart rate icon and the title "Diabetes Risk Assessment". Below the title, it says "Enter your clinical data below to receive an AI-powered diabetes risk prediction and personalised health guidance." A progress bar shows four steps: 1. Personal, 2. Medical, 3. Lifestyle, and 4. Health. The "Personal" step is currently active.

Bottom Screenshot: This screenshot shows the "STEP 1 OF 4" form titled "Personal Information". The subtitle reads "Basic demographic details used to calibrate the prediction model." The form includes the following fields:

- BIOLOGICAL SEX:** Radio buttons for "Female" and "Male". The "Male" button is selected.
- AGE GROUP (select your range):** A dropdown menu showing "45 – 49".
- EDUCATION LEVEL:** A dropdown menu showing "College graduate".
- ANNUAL HOUSEHOLD INCOME:** A dropdown menu showing "\$20,000 – \$24,999".

At the bottom of the form, it says "Step 1 of 4" and a "Continue →" button.

Diabetes Risk Assessment

127.0.0.1:5000

Apps | Gmail | Maps | YouTube

All Bookmarks

STEP 2 OF 4

Medical History

Diagnosed conditions and clinical screening results.

HIGH BLOOD PRESSURE

YesNo

HIGH CHOLESTEROL

YesNo

CHOLESTEROL CHECK (in past 5 years)

YesNo

EVER HAD A STROKE

YesNo

CORONARY HEART DISEASE OR HEART ATTACK

YesNo

← Back

Step 2 of 4

Continue →

Type here to search

29°C Mostly clear

10:46 pm 08/05/2026

Diabetes Risk Assessment

127.0.0.1:5000

Apps | Gmail | Maps | YouTube

All Bookmarks

STEP 4 OF 4

Current Health Status

Your recent physical measurements and self-reported health metrics.

BODY MASS INDEX (BMI) (weight + height²)

70

GENERAL HEALTH (self-rating)

3 — Good

POOR MENTAL HEALTH DAYS (last 30 days, 0–30)

4

POOR PHYSICAL HEALTH DAYS (last 30 days, 0–30)

3

DIFFICULTY WALKING OR CLIMBING STAIRS

YesNo

← Back

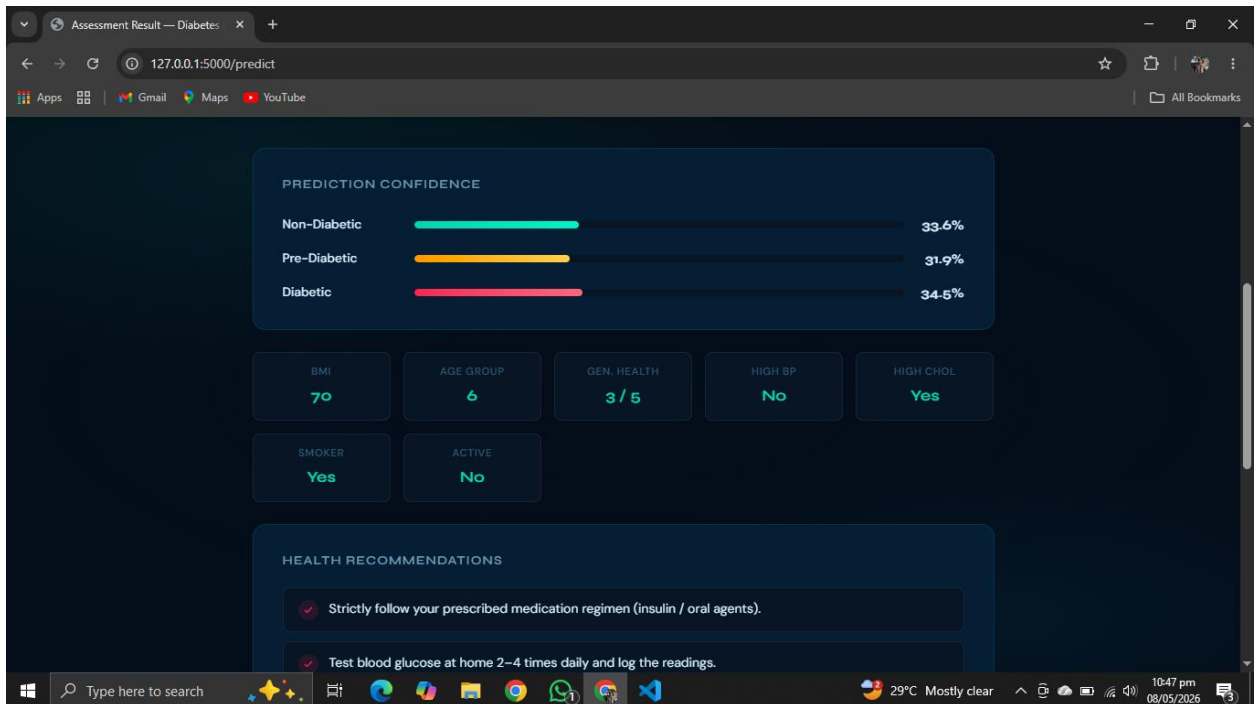
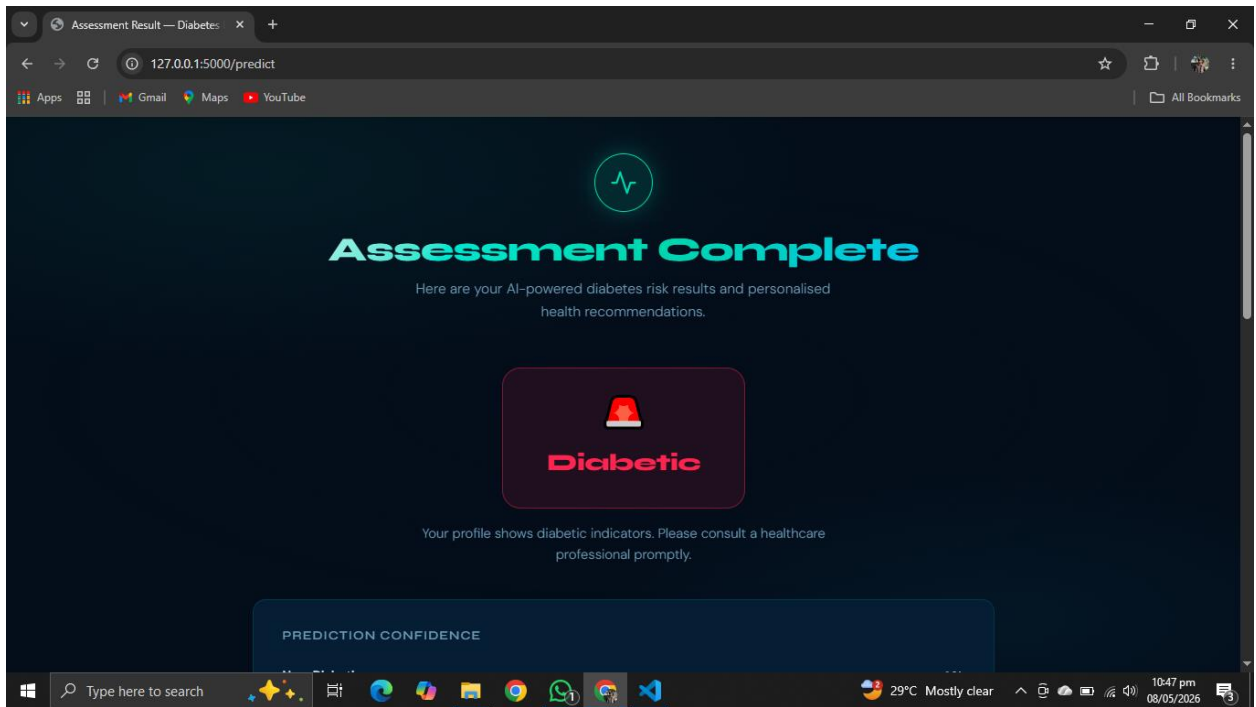
Step 4 of 4

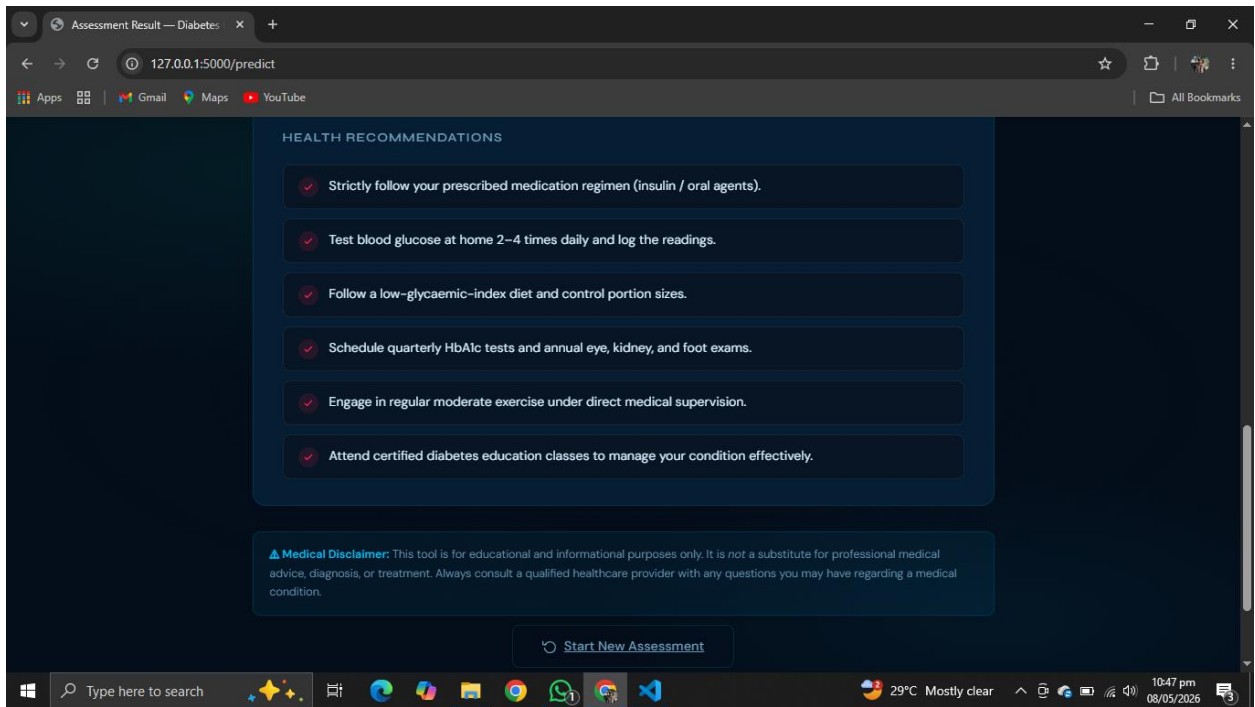
Analyse Risk

Type here to search

29°C Mostly clear

10:47 pm 08/05/2026

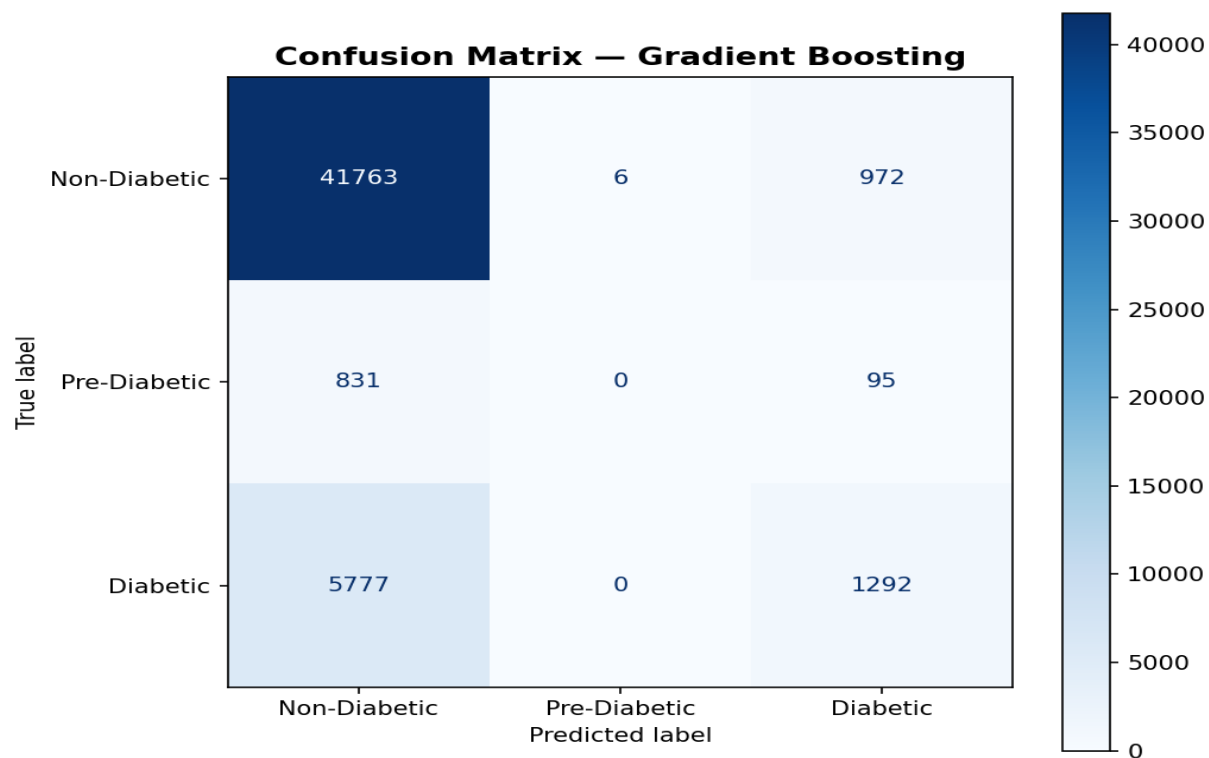
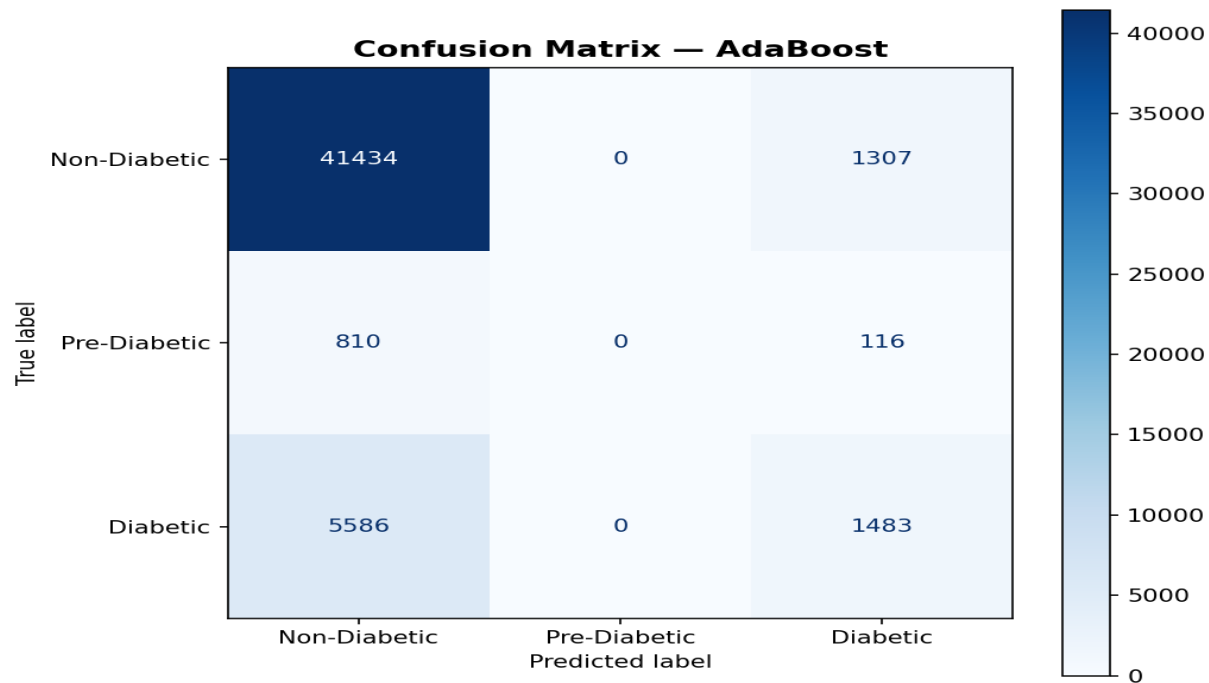


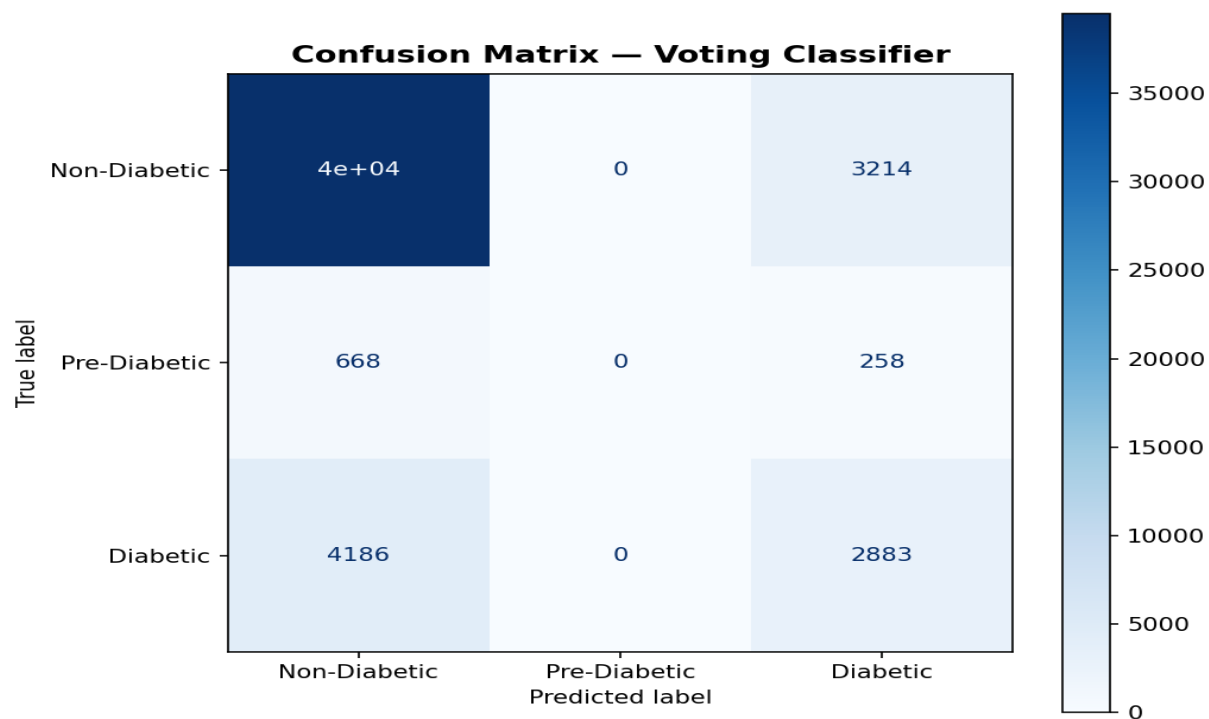
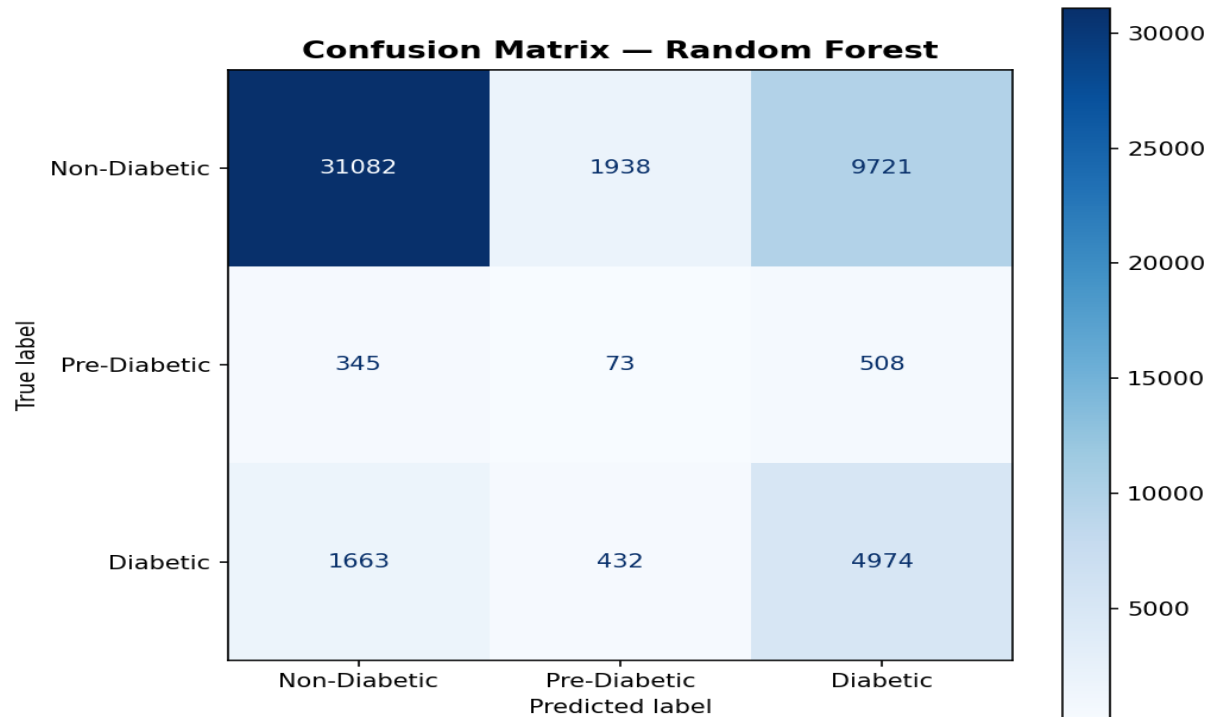


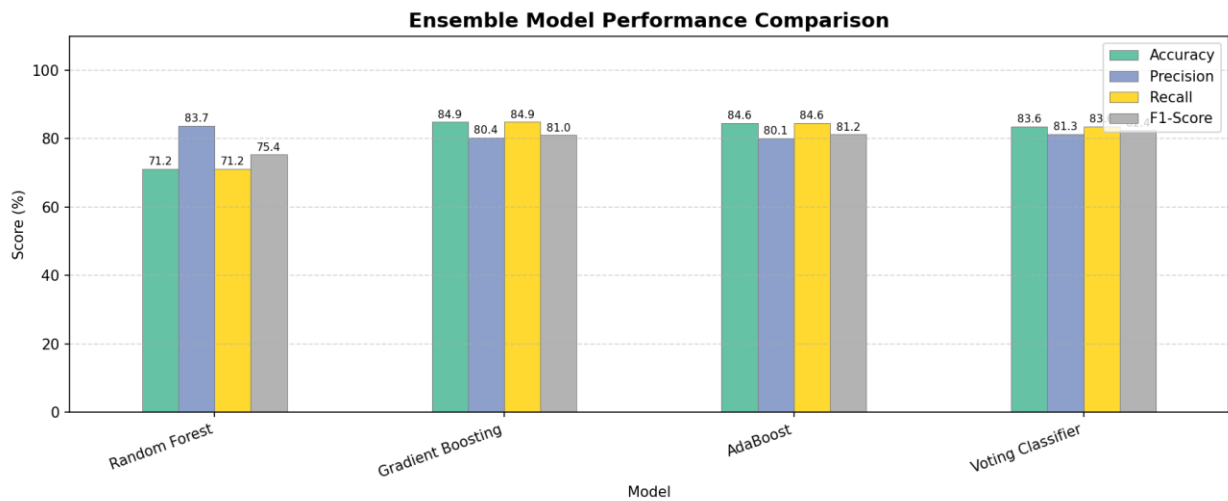
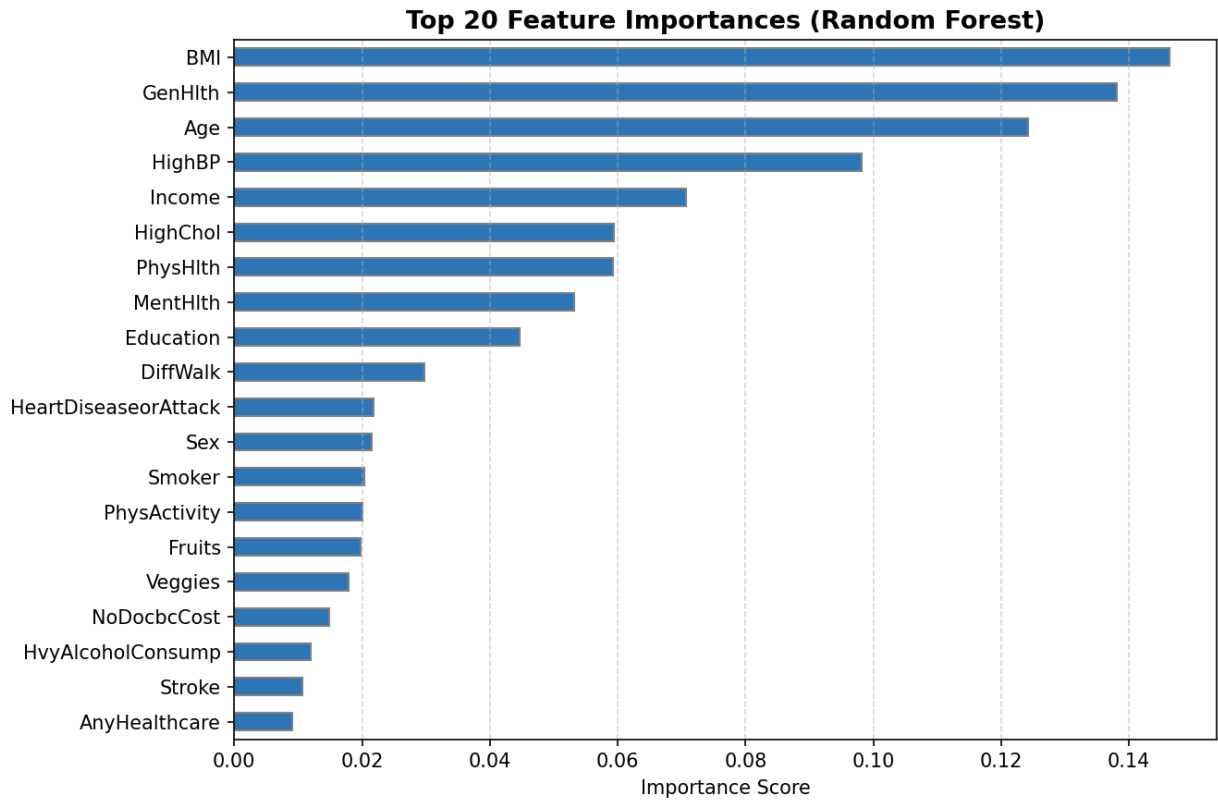
❖ Output plots:

Model Comparison — Accuracy · Precision · Recall · F1-Score (%)

	Accuracy	Precision	Recall	F1-Score
Random Forest	71.21	83.74	71.21	75.36
Gradient Boosting	84.86	80.36	84.86	81.05
AdaBoost	84.59	80.09	84.59	81.22
Voting Classifier	83.59	81.35	83.59	82.43



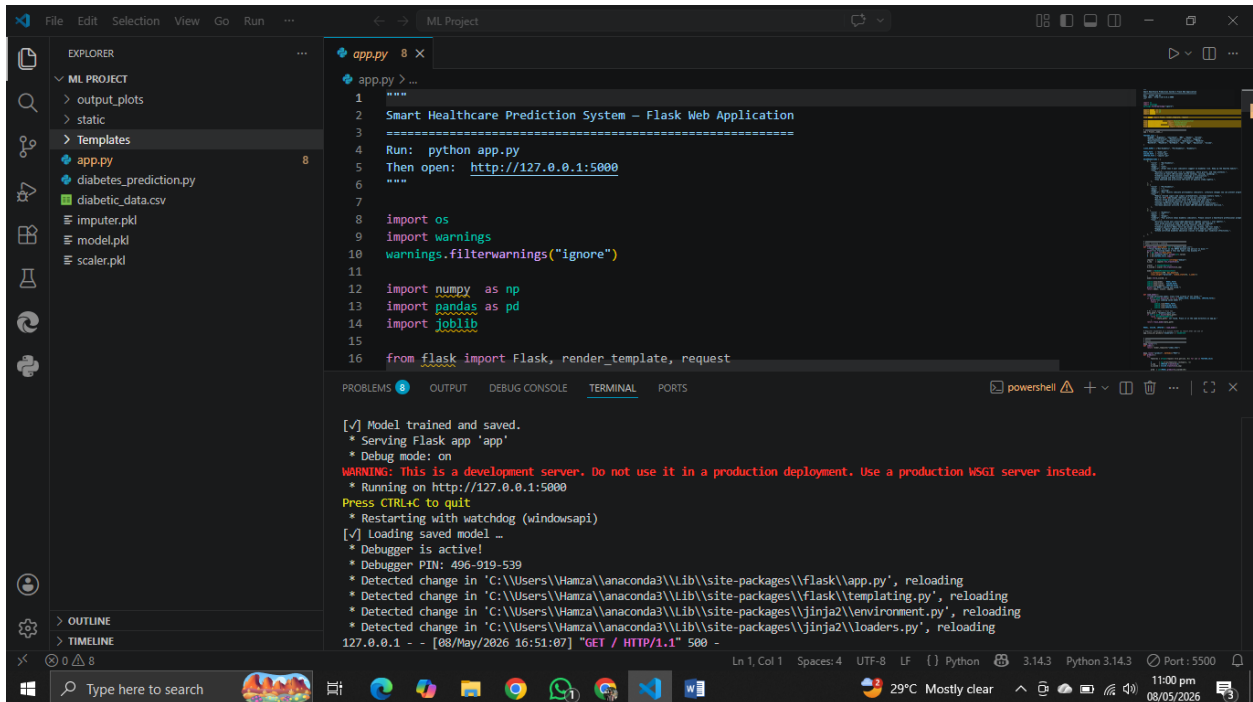




❖ VS Code Screen Shot:

```
PS H:\ML Labs\ML Project> python app.py
[✓] Loading saved model ...
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with watchdog (windowsapi)
[✓] Loading saved model ...
* Debugger is active!
* Debugger PIN: 496-919-539
127.0.0.1 - - [08/May/2026 22:37:26] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [08/May/2026 22:37:26] "GET /static/css/style.css HTTP/1.1" 304 -
127.0.0.1 - - [08/May/2026 22:37:26] "GET /favicon.ico HTTP/1.1" 404 -

Ln 1, Col 1 Spaces: 4 UTF-8 LF {} Python 3.14.3 Python 3.14.3 Port: 5500
```

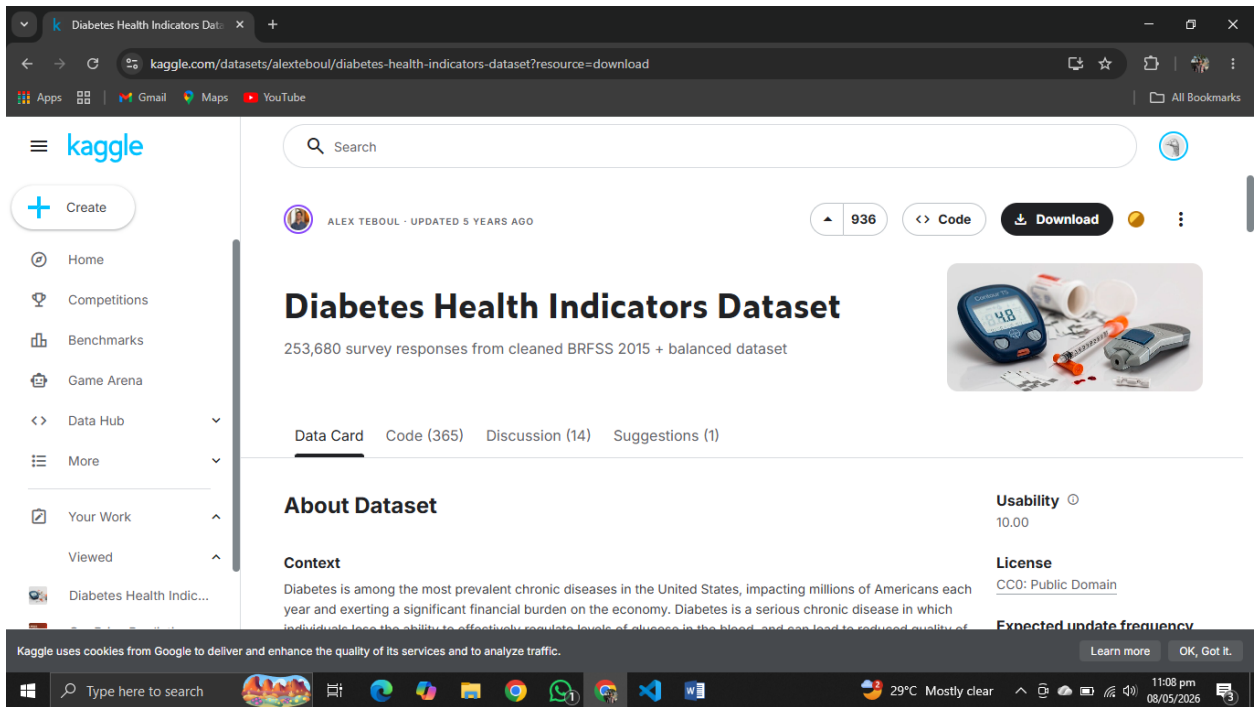


❖ **Dataset:**

Link:

<https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset?resource=download>

Around: 253000 patients real data taken



Rubrics:

Criteria	Excellent (5-6)	Good (3–4)	Needs Improvement (1–2)	Total
Problem understanding, dataset collection, and initial preprocessing	Clear problem definition; dataset thoroughly explored; missing values handled correctly; proper encoding and normalization applied	Problem understood with minor gaps; dataset explored but preprocessing partially complete	Problem unclear; dataset poorly understood; missing values and preprocessing not handled properly	6

Criteria	Excellent (5-6)	Good (3–4)	Needs Improvement (1–2)	Total
Exploratory Data Analysis (EDA) and feature engineering	Comprehensive EDA with visualizations; strong insights; meaningful feature engineering improving model performance	Basic EDA performed; limited visualizations; some feature engineering applied	Little or no EDA; no meaningful insights; feature engineering missing or incorrect	6
Model development (ensemble methods) and evaluation	All ensemble models correctly implemented; metrics (accuracy, precision, recall, F1) properly calculated and well interpreted	Models implemented with minor issues; evaluation metrics partially correct; limited interpretation	Models missing or incorrect; evaluation metrics not calculated or misinterpreted	6
Web deployment, integration, report, and presentation	Fully functional web app; smooth integration; ≥ 4 models compared; clear justification; professional report & presentation	Web app works with minor issues; basic comparison; report/presentation acceptable	Deployment incomplete or not working; no proper comparison; weak or missing report/presentation	6
Total Maks				30